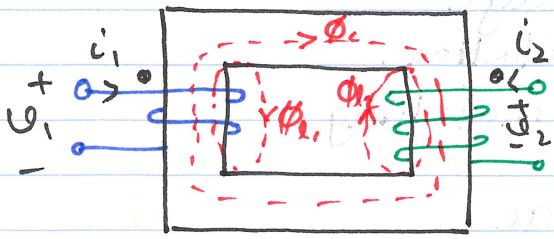
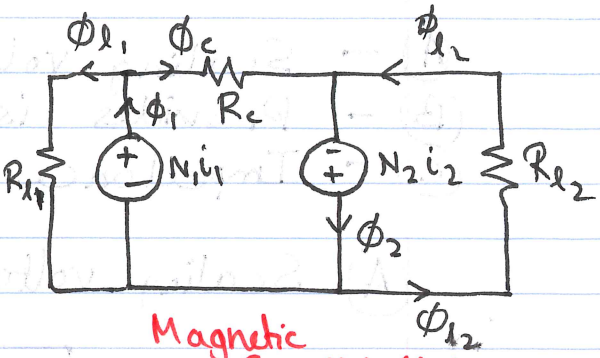
Lecture 13.

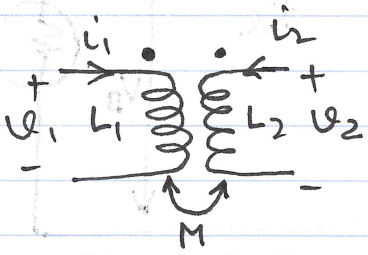
Last time : Coupled coils.



Physical Model



Magnetic Eqv. Ckt. Model.



Electrical Eqv. Ckt. Model

$$V_1 = L_1 \frac{di_1}{dt} + M \frac{di_2}{dt}$$

$$V_2 = M \frac{di_1}{dt} + L_2 \frac{di_2}{dt}.$$

Mathematical model.

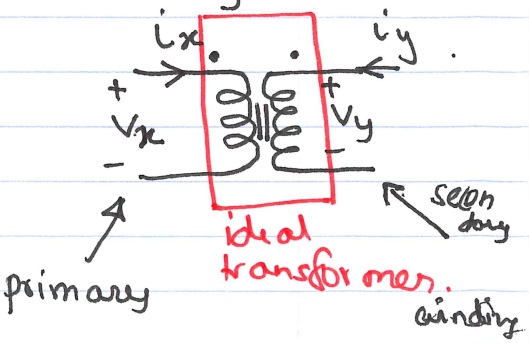
Today : Classic example : Transformer.

If we ignore leakage and consider infinite permeability of the core.

$$N_1 i_x + N_2 i_y = 0 \quad \Rightarrow \quad \frac{i_x}{i_y} = -\frac{N_2}{N_1}$$

and,

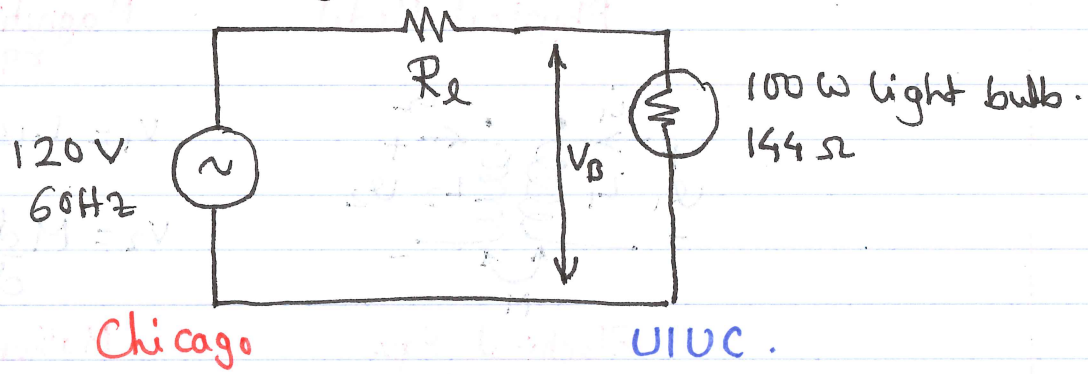
$$\left.\begin{aligned} V_x &= N_1 \frac{d\phi_c}{dt} \\ V_y &= N_2 \frac{d\phi_c}{dt} \end{aligned}\right\} \Rightarrow \frac{V_x}{V_y} = \frac{N_1}{N_2}$$



Why should we care about transformers ?

(A) – Scaling voltage
(B) – Provides isolation.
(C) – Impedance matching.

(A). Scaling voltage



Depending on the diameter of the conductor.

$$R_\ell = \rho \frac{l}{A}$$

ρ → resisting of conductor
l → distance between Chicago and UIUC
A → area of conductor

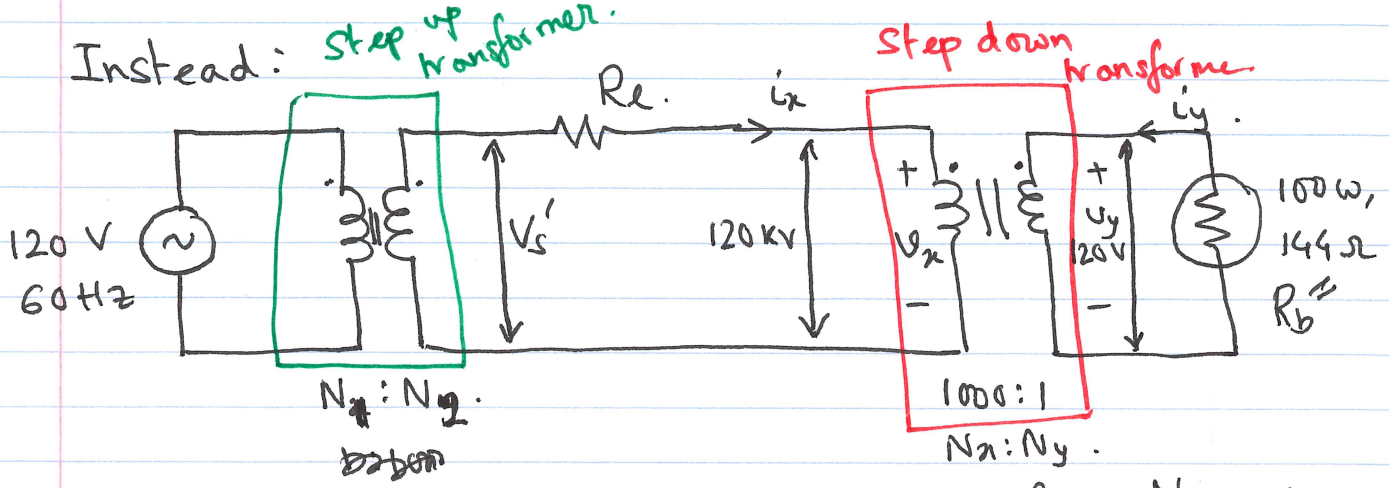
AWG	$R_\ell = \rho \frac{l}{a}$	$I_{rms} = 120/(R_\ell + 144)$	V_B (V)	η (%)
0	71 Ω	$i = 558$ mA	80.4	67%
1	90 Ω	513 mA	73.9	61.6%
⋮				
12	1148 Ω	93 mA	13.4	11.2%

Thickest Cu. gauge ↑ cost

Two big disadvantage : * Voltage drop across the load.
* Efficiency (poor) to transfer power.

All R.M.S quantities here.
for voltage and current.

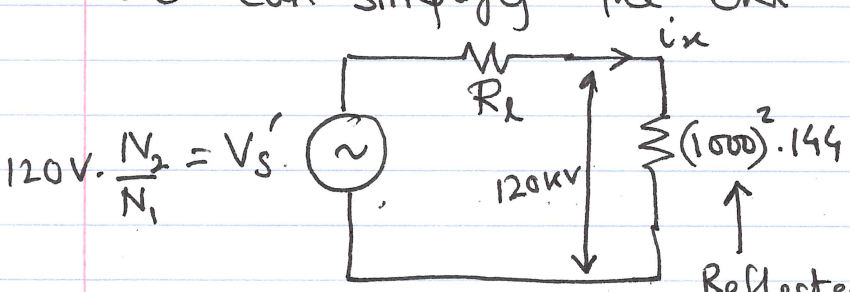
Instead: Step up transformer.



$$v_x = \frac{N_x}{N_y} \cdot v_y$$

$$i_x = -\frac{N_y}{N_x} i_y$$

We can simplify the ckt.



$$\frac{v_x}{i_x} = \left(\frac{N_x}{N_y}\right)^2 \left(\frac{v_y}{-i_y}\right)$$

$$= \left(\frac{N_x}{N_y}\right)^2 \cdot R_b$$

If we choose R_l (12 AWG) = 1148 Ω.

$$i_x = \frac{120\ \text{kV}}{144 \cdot (1000)^2} = 0.833\ \text{mA}.$$

$$\therefore\ V_s' = 120\ \text{kV} + \underbrace{(0.833\ \text{mA}) \times 1148\ \Omega}_{\text{Voltage drop in line.}}$$

$$= 120\text{kV} + 0.956\ \text{V}$$

$$\frac{N_2}{N_1} \approx 1000$$

Power loss in transmission = $i_x^2 \cdot R_l$
= 0.796 mW.

